

Ultrasound-Guided Needle Study Report

Midway Analysis

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Introduction

This study investigates whether holding the needle in the dominant hand results in more efficient performance during an ultrasound-guided needle procedure. Efficiency is measured by the time taken to reach the target within a phantom jelly model.

Twenty novice participants completed ten attempts using each hand arrangement, needle in dominant hand versus needle in non-dominant hand, after receiving standardized instruction. All participants were familiar with ultrasound-guided nerve block procedures but had no prior hands-on experience operating the ultrasound probe or directing needle placement.

The purpose of this report is to analyze the recorded attempt times to determine whether using the needle in the dominant hand leads to faster performance compared to using the non-dominant hand.

Data Description

The cleaned dataset contains 400 observations (20 participants × 20 attempts), with the following variables:

- id** — participant identifier
- hand** — whether the attempt was performed with the needle in the dominant hand (DHP) or non-dominant hand (DHN)
- attempt** — attempt number (1–10 for each hand arrangement)
- time** — time in seconds to reach the target
- DOMINANCE** — participant’s dominant hand (R or L)
- PICKS.UP.PROBE.IN** — which hand the participant naturally used to hold the probe
- STARTING.ARRANGEMENT** — which hand arrangement the participant started with
- SUBJECTIVE.EASIER** — which arrangement the participant felt was easier
- GENDER** — participant gender
- AGE** — participant age
- BACKGROUND** — professional background (e.g., student, nurse)

Methods

The dataset from *info.csv* and *timeval.csv* was merged into a single data frame containing 400 observations. Basic data cleaning included converting categorical variables into factors and verifying that each participant had 10 trials for each hand arrangement.

To explore the distribution of task completion times, a histogram of the *time* variable was produced. A boxplot comparing times between the dominant-hand needle (DHP) and non-dominant-hand needle (DHN) arrangements was also generated to visualize differences in central tendency and variability.

For statistical comparison, the mean time for each hand arrangement was calculated for every participant, resulting in paired observations (DHP mean vs. DHN mean). A paired two-sample *t*-test was performed to determine whether the dominant-hand needle arrangement resulted in significantly faster times.

Results

Figure 1: Distribution of all attempt times

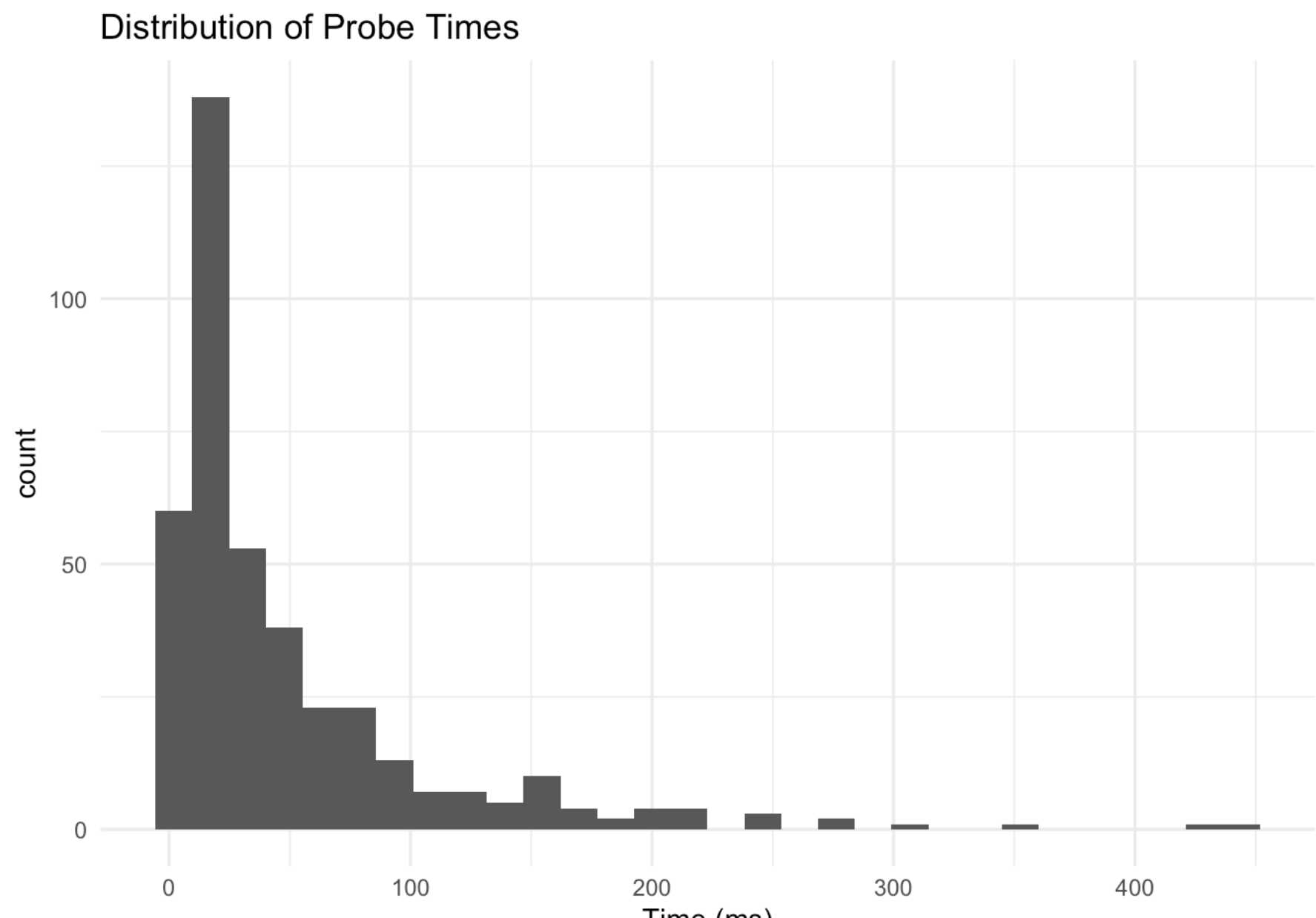


Figure 1 displays the distribution of probe completion times across all participants and trials. The distribution is strongly right-skewed, with the majority of times falling below approximately 60 ms. A long tail extends toward higher values, including several extreme outliers above 200 ms and even beyond 400 ms. This indicates that while most trials were completed quickly, occasional attempts took substantially longer.

Figure 2: Probe Completion Times by Hand Arrangement

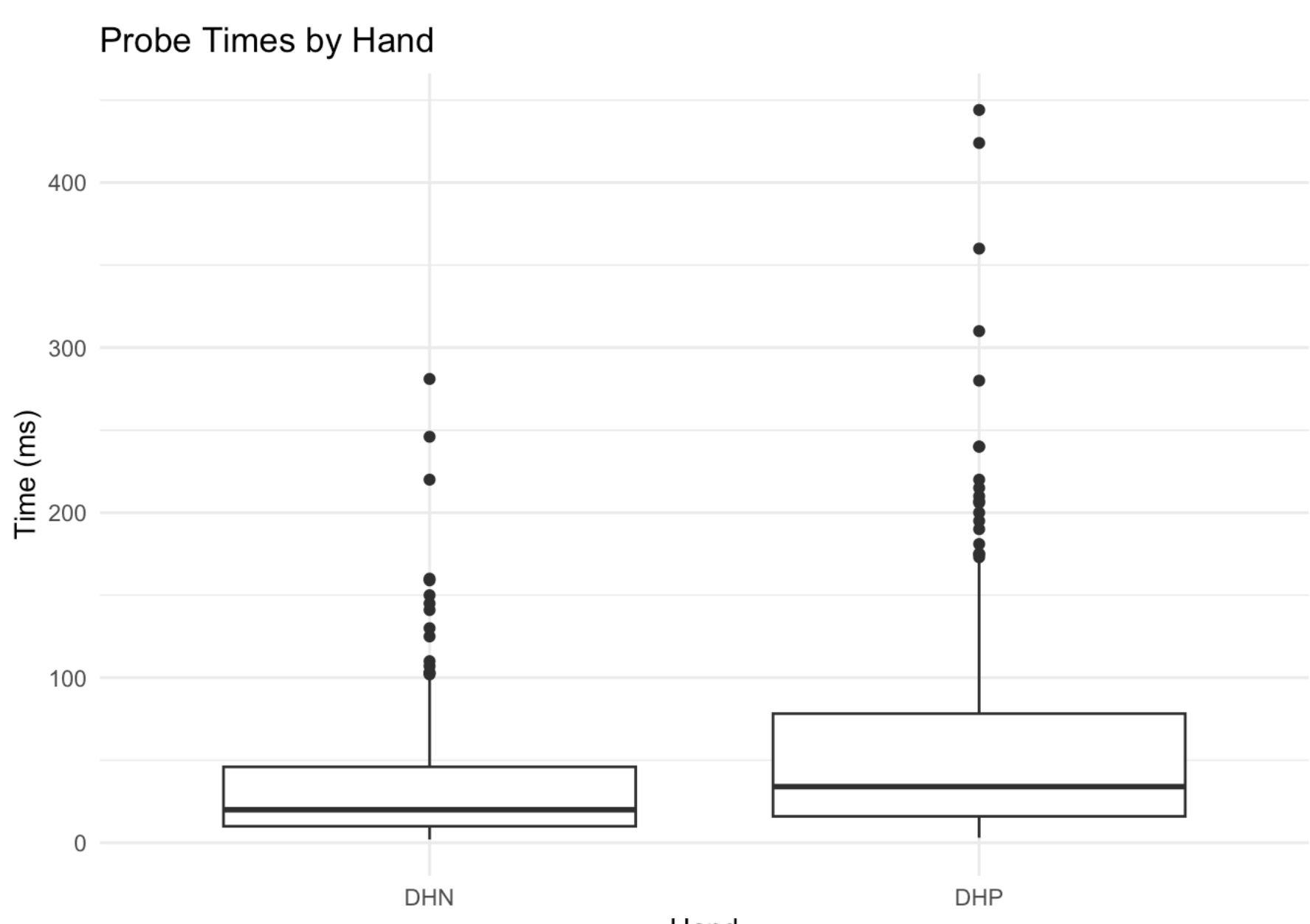


Figure 2 shows the distribution of completion times for the two hand arrangements. The median time is noticeably lower in the DHN condition, and the spread of times is also smaller compared to DHP. The DHP arrangement shows a higher median, larger interquartile range, and more high-value outliers, indicating that participants tended to take longer and were more variable when using the needle in the dominant hand. However, these raw attempt-level differences do not account for repeated trials within participants; therefore, a paired analysis using participant-level averages was conducted to formally compare the two conditions.

Table 1: Paired t-test Comparing DHP and DHN Mean Completion Times

Statistic	Value
t	3.83
df	19
p-value	0.0011
Mean difference (DHP – DHN)	27.37
95% CI	12.40 – 42.34

A paired *t*-test was conducted to compare the average completion times between the dominant-hand needle arrangement (DHP) and the non-dominant-hand needle arrangement (DHN). The results showed a statistically significant difference between the two conditions ($t(19) = 3.83, p = 0.00114$). On average, participants completed the task 27.37 seconds faster using their non-dominant hand. The 95% confidence interval for the mean difference ranged from 12.40 to 42.34 seconds, indicating that the non-dominant-hand arrangement consistently produced shorter completion times.

Summary of Findings

Both the visualizations and the statistical analysis support the conclusion that holding the needle in the non-dominant hand (DHN) results in faster completion times. Figure 1 highlights the overall distribution of times and presence of outliers, Figure 2 demonstrates the consistent speed advantage for DHN, and Table 1 confirms the difference is statistically significant.

Conclusion

The analysis demonstrates that holding the needle in the non-dominant hand (DHN) results in significantly faster completion times compared to using the dominant hand (DHP). On average, participants were 27.37 seconds faster with the non-dominant hand arrangement, and this difference was statistically significant ($t(19) = 3.83, p = 0.00114$). These results suggest that non-dominant hand needle placement may improve efficiency in ultrasound-guided procedures for novice practitioners.

Appendix

A.1 Dataset Preview

Table A1: First 6 Rows of the Probe Times Dataset

	id	hand	attempt	time	DOMINANCE	PICKS.UP.PROBE.IN
	<int>	<chr>	<int>	<int>	<chr>	<chr>
1	1	DHP	1	97	R	NDH
2	1	DHP	2	78	R	NDH
3	1	DHP	3	46	R	NDH
4	1	DHP	4	52	R	NDH
5	1	DHP	5	62	R	NDH
6	1	DHP	6	76	R	NDH

6 rows | 1–7 of 12 columns

A.2 Statistical Test Results

Table A2: Paired t-test Comparing DHP and DHN Mean Completion Times

	id	DHN	DHP
	<int>	<dbl>	<dbl>
	1	99.4	54.9
	2	16.7	14.7
	3	25.6	16.5
	4	28.6	59.3
	5	71.6	118.6
	6	21.4	48.2

6 rows

Paired t-test

```
data: avg_times$DHP and avg_times$DHN
t = 3.826, df = 19, p-value = 0.00114
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 12.39703 42.34297
sample estimates:
mean difference
      27.37
```